RADAR FOR AUTONOMOUS DRIVING

Client	Adrian Boeing
Technical mentor	Fabian Deuser
Prepared by	CITS3200 Professional Computing, Group 07: Aiden Blampain, Damien Zhang, Fariya Zehrin, Fatima Sher, Kelsey Chen, Ricky Yuen
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Evidence snapshot	Project repository, branch main at commit 1704678 (6 October 2026): 41 shared result records, 34 experiment logs
Supersedes	Results walkthrough v2 (30 September 2026)

Repository: *https://github.com/Ricky24248619/Group-7-RADAR-for-Autonomous-Driving*

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1. EXECUTIVE SUMMARY

The report explains the measurements carried out by Group 07 on five datasets (GOOSE, MAN TruckScenes, TruckDrive, STONE and RADIATE) and what each of the results does and does not show. It is based on the project's twenty evidence-linked findings, which are set out below in the form of five main messages. Each of the numbers can be traced back to a saved result record, an experiment log or an evidence file in the project repository.

The headline states that, in TruckDrive with the only dataset we have containing labelled vehicles at a distance of 200 to 400 metres, LiDAR returns were more often found inside the labelled vehicles than radar returns were. We were unable to use this to carry out a detection comparison since no available pretrained model takes input at that range. Consequently, the project's initial hypothesis, that 4D radar would degrade less than LiDAR at long range, is not supported by our geometric evidence and has not yet been tested as a detection issue.

FIVE KEY MESSAGES

1. In the TruckDrive clips, LiDAR provided more information about vehicles at a distance. Across five scenes, out of 504 observations, the labelled vehicles had LiDAR returns in 80.36% of the cases, compared with 37.50% for radar, and in 31 out of 36 observations (radar: 12) in two additional clips. This suggests that LiDAR could be used to obtain early information before a truck needs to respond; however, it is still necessary to measure both the stability of recognition and stopping performance. One caveat is that on TruckScenes, when LiDAR is thinned to the same point count as the radar, its ability to detect vehicles beyond 50 m disappears. This kind of control has not been carried out on TruckDrive (Sections 4.2 and 5).

2. The radar did not provide complete coverage of all important targets. In TruckScenes, for instance, it registered signals in 99.12% of truck observations, 67.66% of observations involving adult pedestrians and 39.00% of those involving traffic cones. Since the samples for the different classes are not the same, these figures represent coverage observations and not evidence of an effect

due to size or material. Separate acceptance checks are required for people and work-zone objects (Section 4.3).

3. For off-road sensing, it is necessary not only to record points but also to correctly interpret the terrain. The GOOSE model that was tested at times regarded hazards as being ground (the number of obstacle points termed ground rises from 2.99% at distances of 0–25 m to 14.17% at 100–150 m) and at other times classified potential ground as an obstacle. Even if a map is constructed from accurately labelled data it may exclude a small number of hazards on the basis of a majority-ground rule, this occurring in 15.88% of the mixed cells (Section 3).

4. A ground label by itself is not sufficient to indicate the space available for a vehicle; within a distance of 25 metres, 43.46% of the half-metre map cells showed no return, obstacle returns were at varying heights, and in order to require support over a 2.5 m square only 23.48% of the positions which had passed a single-cell check were retained. These diagnostics show which information should be preserved for navigation; they do not prove that a clearance or a route exists (Section 3.4).

5. Any future improvements to the model must include tests which are capable of revealing regressions and reflect the way the model is intended to be used. With more model context, six out of the seven held-out terrain frames were improved, one was made worse, and processing time increased by about 28%. A combined soil score of 99.53% masked a rainy-recording result of 7.92%. Water has not yet been tested (as stated in Sections 3.3 and 3.5).

WHAT WE COULD NOT ESTABLISH

• A comparison was carried out with regard to range. Since the pretrained radar/LiDAR pair in question truncates its input at a distance of about 70 m, none of the 504 centre points of the distant vehicles lie within this range. The paper on the TruckDrive dataset states that the current models "do not generalize to ranges beyond 150 metres, with drops between 31% and 99%" (Section 8).

• A weather rating; the evidence of fog consists of a single sequence of four tracks with no clear-weather reference. In the TruckScenes rain and snow recordings (Sections 4.3 and 6) the LiDAR maintained good car coverage.

• The distance needed for safe traversal, clearance, or stopping was not measured.

• The situation in the industry. Our conclusion regarding the long range, along with our conclusion that it is not currently possible to carry out testing for 200–400 m recognition, is in line with the sensor choices that have been published by the industry and also with the TruckDrive authors' own paper. In the cases of weather and model transfer, the views put forward by the industry are in the opposite direction (see Section 10). Section 11 contains the open issues and four decisions on which we should welcome the client's opinion.

2. THE INVESTIGATION CARRIED OUT AND THE WAY TO READ THE RESULTS

The aim of the project is to carry out an evidence-based comparison of the ways in which RADAR, LiDAR and camera systems perceive their environment in the context of autonomous trucking, by using existing open-source models on public datasets which have been recorded in a reproducible manner and then provided as a fully documented body of work. The main research question (D-04) is whether the current models fail beyond a distance of about 150 metres and whether 4D radar degrades less than LiDAR at that range. Each of the datasets was used as a tool for addressing a different aspect of that question.

TABLE 1. DATASETS INVESTIGATED

Dataset	Pair	Question it answers	Sensors used	What we used
GOOSE	Damien Zhang, Ricky Yuen	Can a model separate drivable ground from obstacles off-road?	LiDAR (Velodyne VLS-128) with 64-class human point labels; no labelled radar	961 validation frames characterised; 24 + 7 framesfor model diagnostics
MAN TruckScenes	Aiden Blampain, Fatima Sher	Radar vs LiDAR vs camera on shared ground truth (highway, yards)	Six 4D radars, six LiDARs, cameras	v1.2-mini: all samples for coverage; 80-sample mini_val for models; 389 timing-eligible paired samples
TruckDrive	Fariya Zehrin, Kelsey Chen	Does perception degrade beyond about 150 m?	4D FMCW radar, long-range FMCW LiDAR, cameras	24 mini scenes (radar, labels); camera/LiDAR for 2 scenes; 5 scenes plus 2 hold-out clips for paired support
STONE	Ricky Yuen	Off-road terrain with labelled 4D radar	3 Continental ARS 548 radars and LiDAR	60 frames, 20 from each of 3 recordings
RADIATE	Ricky Yuen	Does radar see what LiDAR misses in fog? LiDAR	Scanning radar and	One public fog sequence: 17 radar frames, 39 vehicle observations, 4 tracks

Six further datasets were reviewed at source or metadata level (Section 7). The rules below apply to every result in this report.

- Coverage is not detection. Most results count whether a sensor return lands inside a labelled object (“geometric support”) or where returns fall by distance. That shows evidence is present, not that a detector would find the object.
- No cross-dataset ranking (decision D-01). Sensors, labels and splits differ between datasets. A segmentation score is never compared with a detection score.
- Denominators are stated, and observations repeat. The same object appears in many frames, so counts are not independent trials.
- Labels favour LiDAR. TruckScenes and TruckDrive boxes were drawn from LiDAR, which favours LiDAR in support counts. RADIATE labels come from radar, which favours radar.
- Every figure states its dataset, sample, sensor, ground-truth basis and limitation.

3. GOOSE: OFF-ROAD TERRAIN (DAMIEN ZHANG, RICKY YUEN)

3.1 WHAT THE DATASET CAN ANSWER

GOOSE is the project's only coreoff-road datasetwith labelled points. Its validation split holds 961 LiDAR frames across eight scenarios, with 174,891,807 points labelled into 64 classes by human annotators. It cannot test long range: across the split, 3.84% of points lie beyond 100 m and 1.10% beyond 150 m. It also has no labelled radar; its Smartmicro radars are not documented as 4D. GOOSE therefore answers the off-road question: can a model tell drivable ground from obstacles?

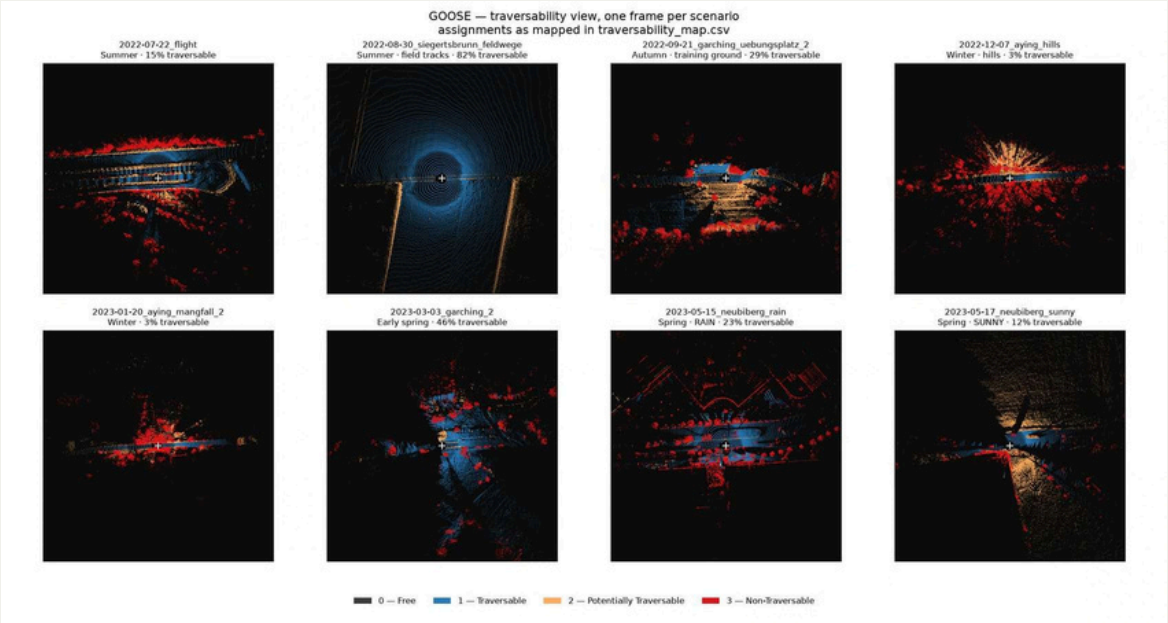


Figure 1. GOOSE traversability grouping, one frame per scenario

Dataset: GOOSE 3D, base validation split
Sample: One frame from each of the 8 validation scenarios
Sensor: LiDAR only (Velodyne VLS-128), bird's-eye view, +100 m around the vehicle
Ground truth/ prediction: Human labels only,nomodel. The four levels (Free, Traversable, Potentially traversable, Non-traversable) are the team's grouping of the 64 classes in traversability_map.csv
Limitation: The grouping is a judgement; 14 contested assignments are documented. “Traversable” is not “safe to drive”: slope, softness and clearance are not labelled

The same human labels look very different by scenario once grouped: 3% of points are traversable in the winter-hills frame and 82% on the summer field tracks. This is not a model deciding where to drive.

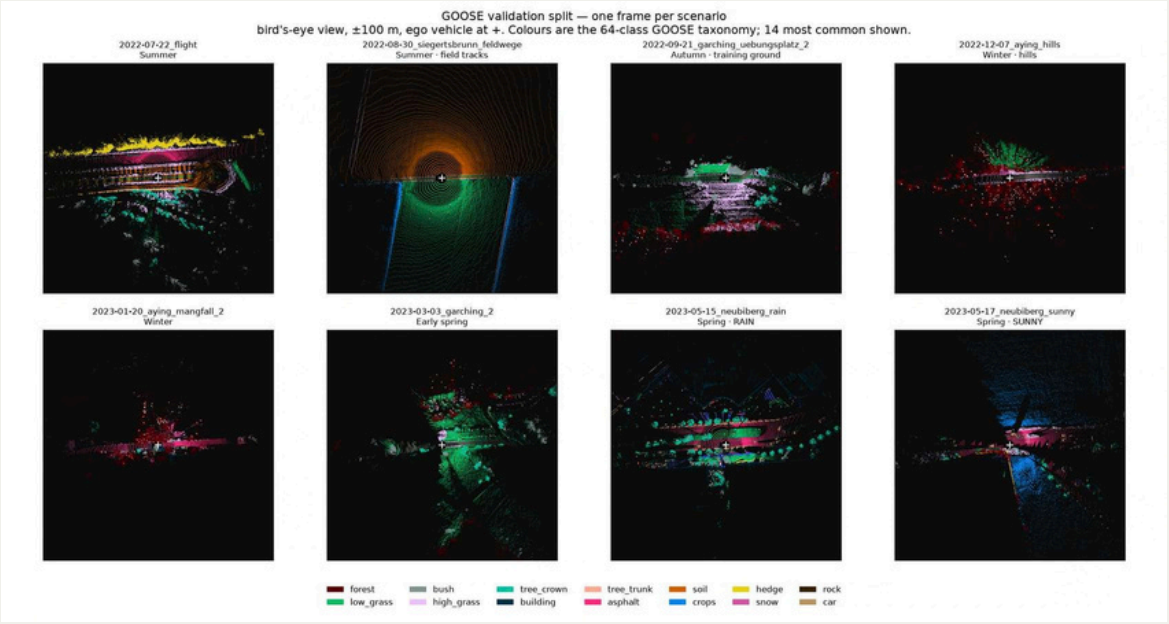


Figure 2. GOOSE scene variety in the 64-class labels

Dataset: GOOSE3D, base validation split (961frames,174,891,807 labelled points)
Sample: One frame from each of the 8 scenarios, across summer, autumn, winter and spring, including rain
Sensor: LiDAR only, bird's-eyeview,+100m
Ground truth / prediction: Human labels, coloured by the 64-class taxonomy (the 14 most common appear in the legend).No model
Limitation: Shows variety only, one frame per scenario; not statistics or model performance

3.2 WHAT A PUBLISHED MODEL DOES ON GOOSE

The published PTv3 segmentation model could not run on Apple Silicon (no CUDA path). On the team's one CUDA device, a GTX 1660 with 6 GiB, it runs only in FP32 with an attention patch of 64 instead of the published 1,024. The full validation run was stopped after 10 of 961 frames at 18.94 s per frame. Our numbers are therefore diagnostics of a reduced configuration, never a reproduction of the published 0.8096 mIoU.

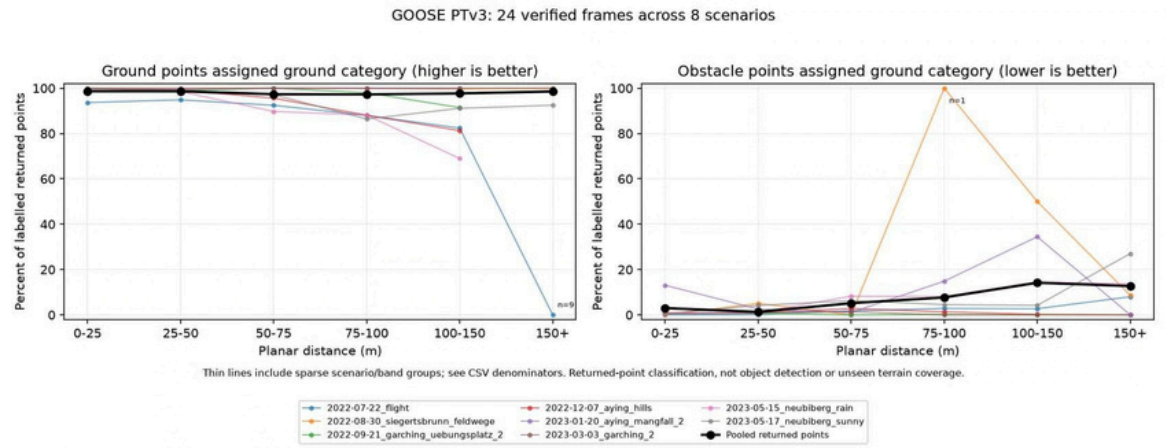


Figure 3. What the reduced PTv3 model mistakes for ground, by distance

Dataset: GOOSE3D, base validation split
Sample: 24 preselected frames, three from each ofthe8scenarios(4,192,081labelled points)
Sensor: LiDARonly
Ground truth / prediction: PTv3 predictions scored against human labels, regrouped into ground and obstacle categories
Limitation: Reduced configuration (patch 64, FP32).Thinlinesare singlescenarios and some bandsarevery sparse (forexample n=1). Points arecorrelated,not independentobjects. Segmentation, not detection

Ground points are assigned the ground category 97–99% of the time over all distance bands. The true failure is the opposite: obstacle points classified as ground rise from 2.99% at 0–25 m to 14.17% at 100–150 m. An earlier 10-frame result, where ground recognition decreased with distance, did not generalize across scenarios, since 85% of the ground points at 100–150 m came from one easy scenario.

The errors come from a few objects. In the error-case study, 96.92% of nearby rock-to-ground errors come from one frame and three rock instances, and 98.75% of nearby building errors come from the same frame. One must never interpret a statement such as “40% of rock points confused” to mean “40% of rocks missed”.

Part of the reason is our reduced configuration. On three fixed frames, increasing the attention patch from 64 to 128 to 256 decreased nearby rock-called-ground from 49.52% to 25.02% to 22.56%, and building-called-ground from 18.43% to 10.18% to 4.12%. A repeat of the baseline produced identical predictions, so this is not noise. It is not a complete solution, however, as many rocks are simply relabeled to a different incorrect category.

3.3 HELD-OUT FRAMES: MORE CONTEXT HELPS SOME RECORDINGS AND HARMS OTHERS

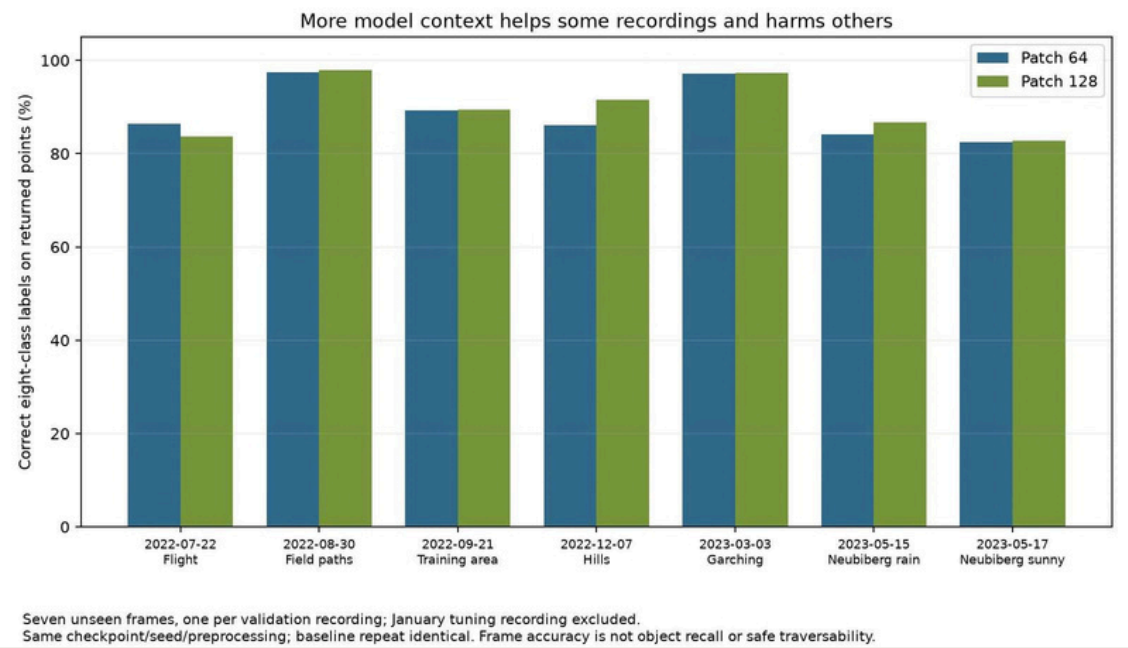


Figure 4.Per-recording effect of more model context on held-out frames

Dataset: GOOSE 3D, base validation split
Sample: Seven unseen frames, one per validation recording (the January tuning recording is excluded)
Sensor: LiDAR only
Ground truth/ prediction: Share of returned points given the correct eight-class label, attention patch 64 vs 128; same checkpoint, seed and preprocessing
Limitation: One selected frame per recording. Frame accuracy is not object recall or safe traversability

Context 128 improved overall point-category correctness in six of the seven frames and worsened the flight frame from 86.37% to 83.69%. Summed logged batch time rose from 93.25 s to 119.22 s, about 28%. Three further results show why per-class and per-recording reporting matters:

- Tall grass: at patch 128, 48.94% of tall-grass points across the seven frames were called ground.
- Ground is not the same as the right ground. Gravel had 99.82% of points called ground but only 80.07% in the correct natural- or artificial-ground category; asphalt had 99.42% and 89.78%.
- Pooled scores hide weak recordings. Pooled soil correctness was 99.53%, dominated by field-path points. In the selected rainy frame only 7.92% of 240 soil points had the correct broad category, although 224 of 240 were still called ground. This is one small frame and does not establish a causal rain penalty.

3.4 NAVIGATION-RELEVANT INFORMATION THAT A GROUND LABEL LEAVES OUT

A planner needs more thana ground/obstacle label. The diagnostics below use the 31 selected GOOSE frames (the original 24 plus the seven held-out frames) and, where stated, the correct human labels, so they isolate what is lost in the conversion from labels to a map.

TABLE 2. DIAGNOSTICS ON 31 GOOSE FRAMES

Question	What we measured	Reading
How much nearby space is unobserved?	105,904 of 243,660half-metre cell observations within25 m had no return (43.46%); at 1 m cells, 16,654 of 61,256 (27.19%)	Keep unknown space explicit; a gap must not silently become ground
Can a map hide a hazard even with correct labels?	Of 4,232 half-meter cells holding both firm ground and a rigid, person or vehicle hazard, 672 (15.88%) had a ground majority	A majority-ground rule would erase the minority hazard
Does the model also reject usable ground?	2,853 of 1,312,878 candidate-firm-surface points were given a blocked category (patch 64)	Track both error directions in acceptance tests
Do obstacles sit at one height?	Of 609,511 rigid-hazard points within 25 m, 242,801 had a ground reference within 1 m; of those, 64,920 sat at least 2 m above it and 160,235 sat 0.2–2 m above it	A flat map loses vertical structure that matters for clearance
Is labelled ground smooth?	28,958 of 70,807 half-metre ground cells met the plane-fit criteria (RMS ⩽2 cm in 20,920; 2–5 cm in 7,244; ›5 cm in 794); 20,927 had fewer than five points	A ground label is not a smoothness or traction guarantee
Does a wider footprint change the answer?	Of 61,210 centre cells with ground and no hazard, complete support for a 1.5 m square kept 23,598 (38.55%) and for a 2.5 m square kept 14,371 (23.48%)	Evaluate the vehicle's actual footprint, not a single cell

Limit: these are diagnostic and oracle-label analyses on selected frames. Cells and points are repeated observations, not independent square metres or objects. Nothing here certifies clearance, traction, grade or a usable route.

3.5 INDEPENDENT TERRAIN DATA

A transfer test on independent data was prepared but not run. Five preselected WildScenes frames contain 1,176 log points, 24 rock points, one mud point, and no water points. The 31 selected GOOSE frames contain no water either, and the WildScenes benchmark maps water to its ignored label. WildScenes' labelled clouds decode as XYZ only, while the GOOSE model's input uses XYZ and intensity, and its grass classes lackGOOSE's short/tall distinction. RELLIS-3D, which has separate water, puddle, mud, and rubble classes, was audited at the metadata level only. Water performance is untested in this project.

Sources: *evidence/GOOSE/README.md;docs/final-findings-oct03.md (D05–D08, D12–D15, D18–D21); docs/goose-multiscenario-terrain.md; docs/goose-error-cases.md; docs/goose-failure-causes.md; docs/temporal-terrain-findings-oct03.md; result records 0001, 0004–0006, 0026, 0027, 0032–0034, 0037–0039.*

4. MAN TRUCKSCENES: HIGHWAY AND YARDS (AIDEN BLAMPAIN, FATIMA SHER)

4.1 WHAT THE DATASET CAN ANSWER

TruckScenes carries six 4D radars, six LiDARs, and cameras on one truck, with shared 3D box labels, which makes it the project's best-matched sensor suite. It cannot test at 200–400 m. Recorded radar in the mini release stops at about 189.52 m across all 2,400 radar keyframes, and the stock evaluator scores objects only to 75 m or 150 m depending on class. All six radar channels agree on that boundary to 0.000003 m across 1,145,496 returns. That suggests a shared acquisition or processing limit, but the cause is unknown and not an established sensor specification.

4.2 WHICH SENSORS PUT EVIDENCE ON LABELLED OBJECTS, AND WHY

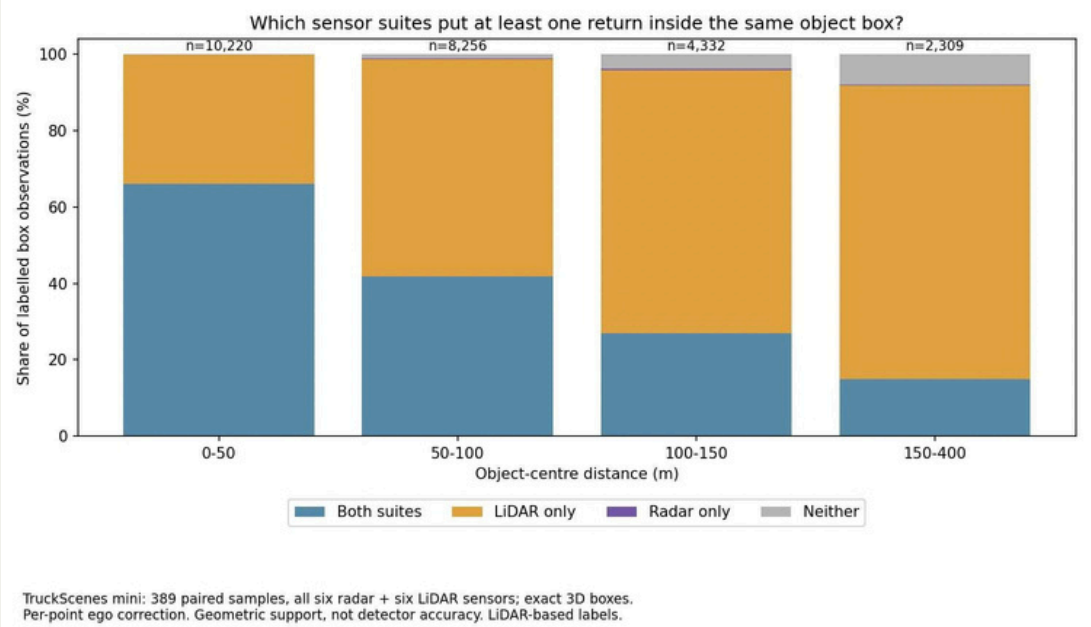


Figure 5.Which sensor suites put at leastone return insidethesame object box

Dataset: TruckScenes v1.2-mini
Sample: 389 paired samples after timing exclusions; 25,117 labelled box observations
Sensor: All six radars and all six LiDARs, with per-point ego-motion correction
Ground truth/ prediction: For each labelled 3D box: did at least one radar and/or LiDAR return land inside it? No model
Limitation: Geometric support, not detection accuracy. Labels were drawn from LiDAR, which favours LiDAR. Observations repeat across frames

LiDAR puts a return inside far more boxes than radar at every distance. At 150–400 m, LiDAR supports 2,118 of 2,309 observations and radar 350, and radar almost never sees an object that LiDAR misses. Timing matters here: correcting individual LiDAR point times cut apparent radar-only observations from 185 to 39.

How much of that gap is simply point count? The LiDAR cloud holds a median of 77.9 returns for every radar return. Fatima's control (record 0041) keeps, in each frame, a random subset of LiDAR returns the same size as that frame's radar returns, then recounts support exactly (Figure 6).

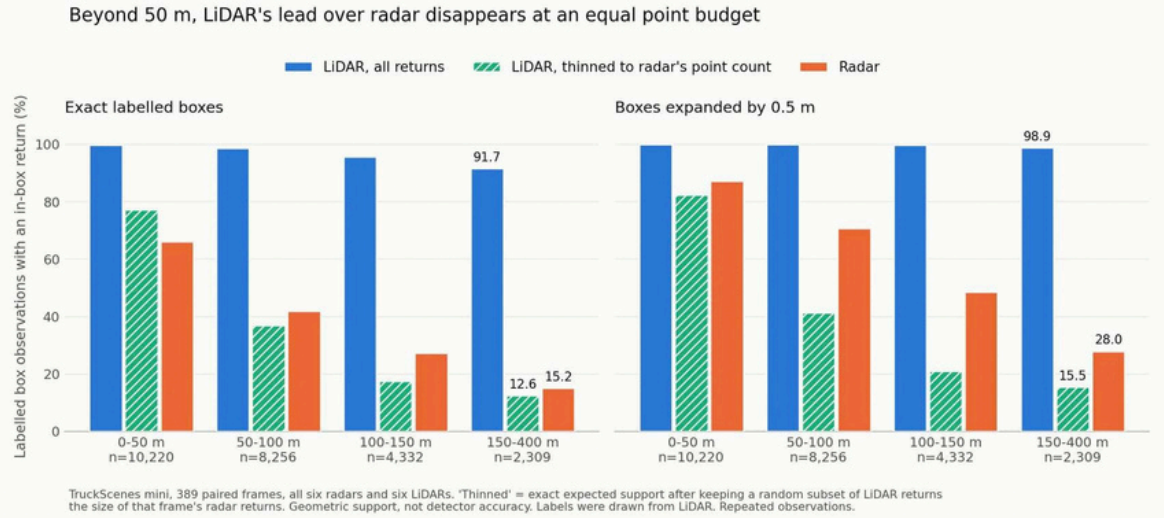


Figure 6. LiDAR support at an equal point budget, by distance and box margin

Dataset: TruckScenes v1.2-mini
Sample: The same 389 frames and box observations as Figure 5; n per band shown under each group
Sensor: All six LiDARs (full, and thinned to the radar's point count) and all six radars
Ground truth / prediction: Exact expected support after uniform random thinning (hypergeometric), consistent with 200 seeded draws. No model.
Limitation: Random thinning is one null model; a lower-resolution sensor would thin structurally. It bounds how much arithmetic explains, not how a cheaper LiDAR would behave. Chart drawn for this report from the committed band_summary.csv

BEYOND 50 M, LIDAR'S SUPPORT ADVANTAGE DOES NOT SURVIVE AN EQUAL POINT BUDGET, AT EITHER BOX MARGIN.

At 150–400 m with exact boxes, thinned LiDAR supports 12.6% of observations against radar's 15.2%, down from 91.7% for the full cloud. With boxes expanded by 0.5 m it is 15.5% against radar's 28.0%. Most of the long-range gap in Figure 5 is therefore explained by point count alone. Below 50 m the result depends on the margin: thinned LiDAR still leads radar with exact boxes (7,901.5 against 6,753 supported observations) but not with expanded boxes (8,425.2 against 8,922). That close-range advantage is not robust, so it is not reported as a finding. Counting returns by distance is easy to misread, as Figures 7 and 8 show.

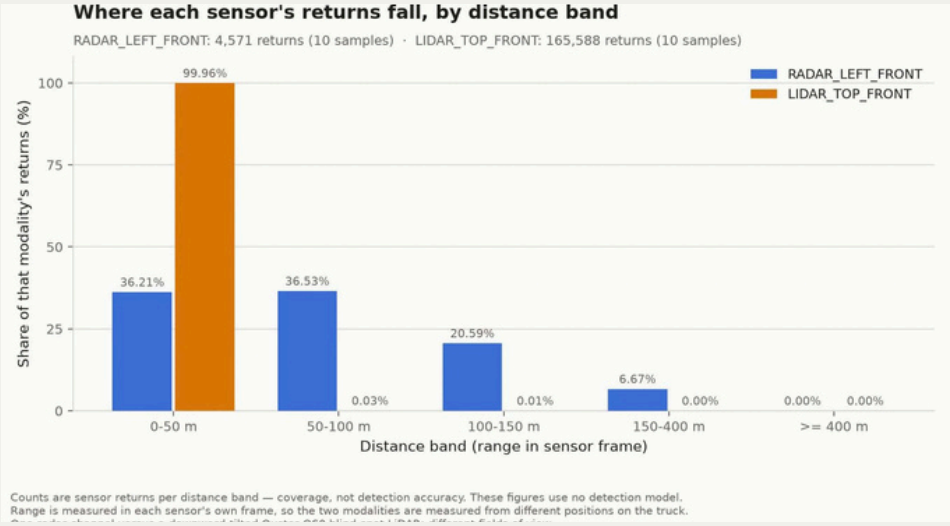


Figure 7. Where one radar's and one blind-spot LiDAR's returns fall, by distance

Dataset: TruckScenes v1.2-mini

Sample: 10 declared samples, the first annotated sample of each mini scene

Sensor: RADAR_LEFT_FRONT (4,571 returns) vs LIDAR_TOP_FRONT (165,588 returns), each measured in its own sensor frame. These are not matched.

Ground truth / prediction: Return counts and shares by distance band. No labels, no predictions.

Limitation: LIDAR_TOP_FRONT is a roof Ouster OS0 rated 35 m and angled down for blindspot coverage, so this reflects its mounting, not LiDAR range in general. Coverage, not detection.

Radar returns spread out to 189.46 m, while this particular LiDAR keeps 99.96% of its returns within 50 m (furthest 140.25 m), as a downward-tilted short-range sensor should. This must not be read as radar out-ranging LiDAR. Figure 8 is the like-for-like comparison.

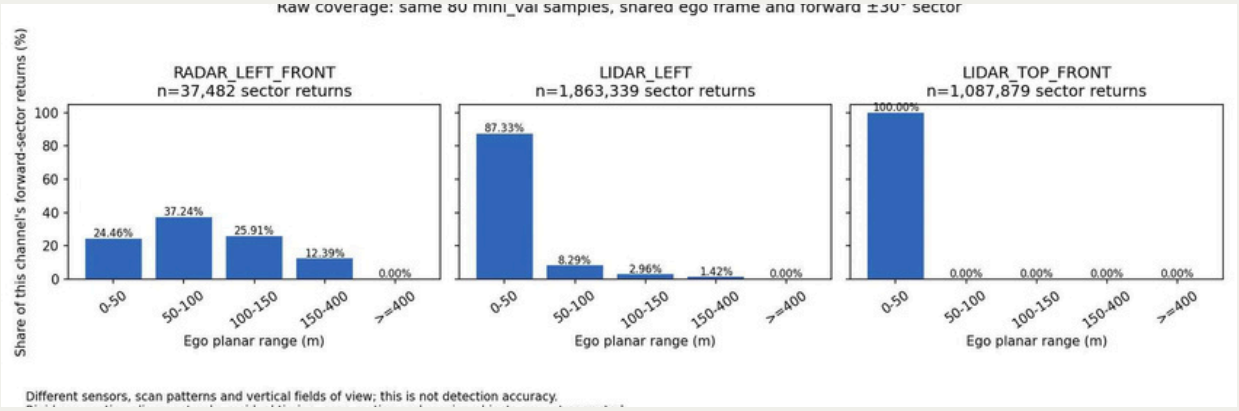


Figure 8. Like-for-like return coverage in a shared frame and sector

Dataset: TruckScenes v1.2-mini, mini_val

Sample: All 80 mini_val samples

Sensor: RADAR_LEFT_FRONT, LIDAR_LEFT (Hesai Pandar64, rated 200 m) and LIDAR_TOP_FRONT, in one shared ego frame, restricted to the same forward +30° sector

Ground truth / prediction: Return shares by distance band. No labels, no predictions

Limitation: Rigid ego alignment only (residual timing and scan motion). Different scan patterns and vertical fields of view. Coverage, not detection

With a suitable LiDAR the picture reverses. At 150 m and beyond, LIDAR_LEFT has 26,367 returns against the radar's 4,643, even though it concentrates 87.33% of its returns within 50 m.

4.3 OBJECT TYPES, WEATHER AND CAMERA VISIBILITY

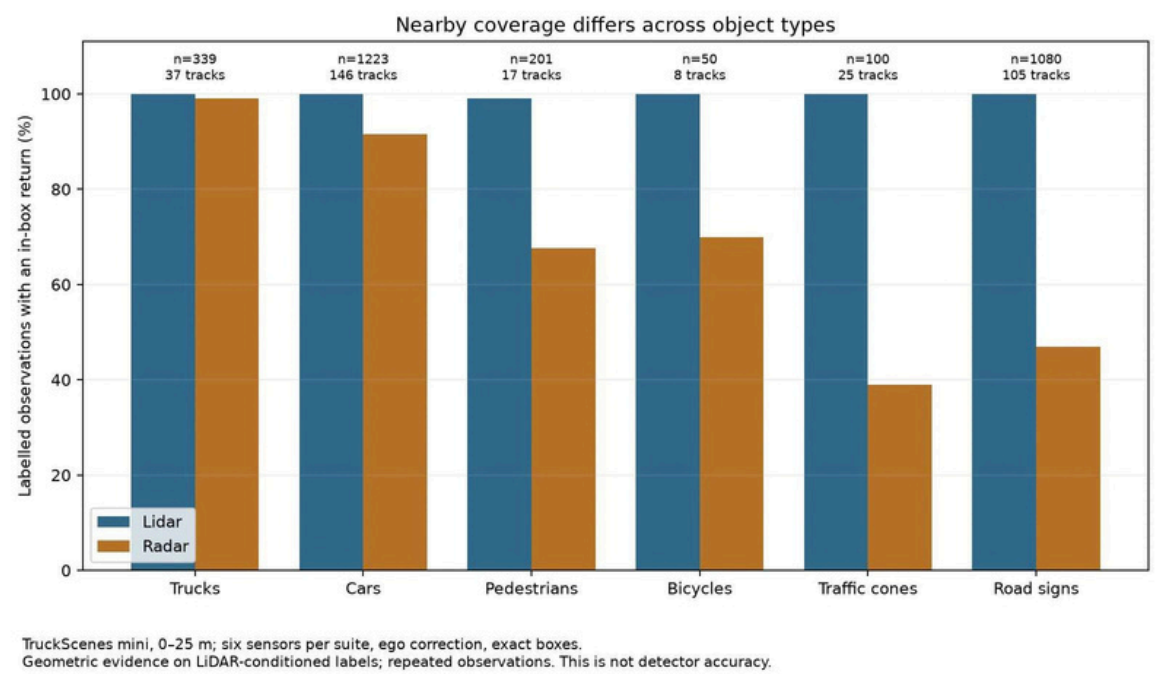


Figure 9.Nearby coverage differs across object types

Dataset: TruckScenes v1.2-mini
Sample: Labelled observations within 0–25 m; n and track counts shown above each pair
Sensor: All six LiDARs and all six radars, ego-motion corrected, exact boxes
Ground truth / prediction: Share of labelled observations with at least one in-box return. No model
Limitation: Class cohorts differ and repeat; labels are LiDAR-conditioned. This does not isolate size, material or visibility as the cause. Not detector accuracy

Radar coverage is near-complete for trucks(3360f339,99.12%) but much less for other classes: adult pedestrians (136 of 201, 67.66%), bicycles(70.00%), road signs (46.94%), and traffic cones (39 of 100, 39.00%). LiDAR is at 99–100% in every category shown. Acceptance tests should include people, bikes, and work-zone objects in addition to the truck-dominated average.

Weather. In the 50–100 m automotive sweeps, the rain record had 776 observations with LiDAR support 95.62% vs. radar 42.40%; the snow record had 498 observations with LiDAR 99.40% and radar 60.84%. Since each condition is a single recording with different targets, this is not a controlled comparison of sensor performance in different weather, but it does not show LiDAR to be ineffective in rain or snow.

Camera visibility. Of 8,125 exact-box observations scored 0–40% visible to the cameras, 95.14% had at least one LiDAR return and 27.03% at least one radar return. Poor camera visibility did not necessarily imply a lack of LiDAR evidence.

4.4 MODEL RESULTS

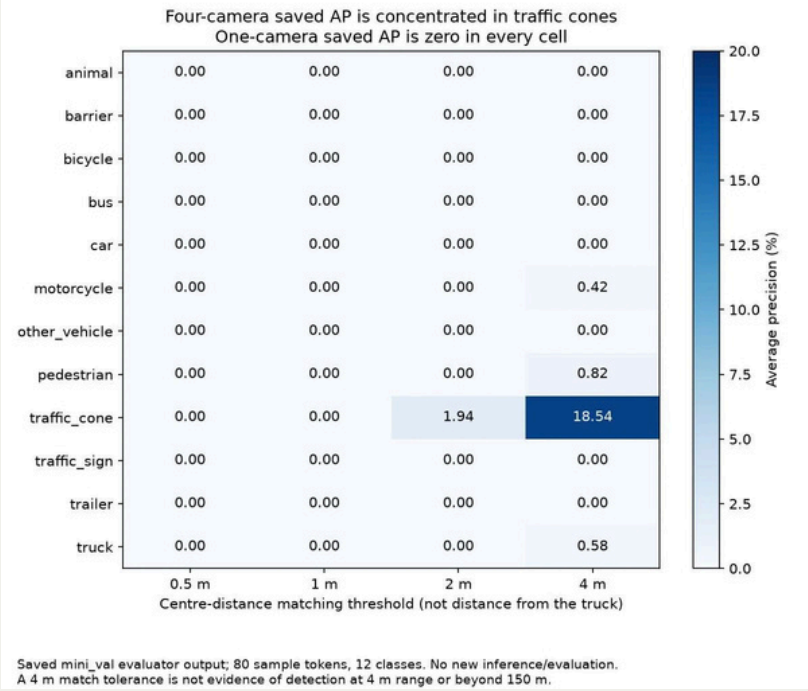


Figure 10. Camera detector (FCOS3D, zero-shot): average precision by class and matching threshold

Dataset: TruckScenes v1.2-mini, mini_val
Sample: All 80 mini_val samples, four cameras.
Sensor: Camera only. FCOS3D pretrained on nuScenes and run without any TruckScenes training.
Ground truth / prediction: Predicted boxes scored against human-labelled boxes.
Limitation: The 0.5–4 m axis is the tolerance between predicted and true box centres, not distance from the truck. The cause of the low score (camera height and pitch, or the yard domain) is not isolated

Four-camera mAP is 0.0046, almost all of it from traffic cones (18.54% AP at the loosest 4 m tolerance); a one-camera run scored 0.0000. A general-purpose camera detector does not transfer to TruckScenes out of the box.

LiDAR detector (PointPillars, zero-shot). The first full run scored mAP 0.0000, but review found the input was wrong: LIDAR_TOP_FRONT is pitched about 56° while the nuScenes checkpoint expects an upright LiDAR, and all 596 output boxes came out tilted 54.6–58.2°. With each cloud rotated into an upright virtual LiDAR, every output box is upright. The model was then run on each LiDAR separately and on all six merged (Table 3).

TABLE 3. POINTPILLARS ON THE 80 MINI_VAL SAMPLES, AFTER INPUT RECTIFICATION

LiDAR input	Mounting	Scored boxes with ⩾1 point	mAP	Car AP	Truck AP
All six merged	Combined	94.9% (1,981 / 2,088)	0.1005	0.406	0.009
LIDAR_LEFT	Side, level	69.5%	0.0555	0.202	0.012
LIDAR_RIGHT	Side, level	67.6%	0.0466	0.239	0.009
LIDAR_TOP_LEFT	Roof, tilted ~55°	31.4%	0.0125	–	–
LIDAR_TOP_RIGHT	Roof, tilted ~55°	42.6%	0.0107	–	–
LIDAR_TOP_FRONT	Roof, tilted ~56° (blind spot)	23.9%	0.0067	0.000	0.017
LIDAR_REAR	Rear, level	23.2%	0.0057	–	–

–: per-class AP not reported for these channels in EXP-0024.

Channel choice dominates single-LiDAR scores, and merging all six nearly doubles the best single channel (mAP 0.1005, car AP 0.406, traffic cone 0.260).

Trucks are a recognition problem, not a visibility problem: the merged cloud has points on all 392 scored trucks and 407 of 408 trailers, yet truck AP is 0.009 and trailer AP 0.110. That points to the nuScenes checkpoint not recognising TruckScenes' heavy trucks and trailers. Traffic signs and animals have no nuScenes class and can never score. Camera and LiDAR results come from different architectures and training, so the gap between 0.0046 and 0.1005 is not a sensor comparison. Both are below 150 m and say nothing about D-04.

Sources: *docs/client-comparison-brief.md; TruckScenes - Fatima/RANGE-BANDS.md; docs/truckscenes-raw-comparison.md; docs/long-range-evidence-audit.md; experiment-log/0010, 0024, 0033, 0034; docs/final-findings-oct03.md (D02, D04, D09, D22); result records 0011, 0012, 0016, 0017, 0029, 0040, 0041.*

5. TRUCKDRIVE: LONG-RANGE HIGHWAY (FARIYA ZEHRIN, KELSEY CHEN)

5.1 WHAT WE HOLD

TruckDrive is the only dataset in our project where vehicles are labelled out to a distance of 400 m. The truck includes seven long-range FMCW LiDARs, as well as ten 4D FMCW radars. Our local mini set includes all 24 scenes for radar, label, calibration and poses (12,502 radar frames, 4,800 box annotation frames and 526,148 3D boxes in 27 classes). Camera and LiDAR were only included for two representative scenes (scene_28_1 and scene_28_22) due to storage constraints, and so no claims cover camera or LiDAR for all 24 scenes.

On our one-frame-per-scene radar scan, we found 6,746 of a total of 71,206 returns (9.47%) that are located 150 m or further away. Note that this count applies only to those 24 selected frames, not the full 12,502 radar frames. The paired radar/LiDAR information for scene_28_1 includes 200 paired frames and 8,113 valid object observations, where labelled objects range from 3.89 m to 399.03 m.

5.2 Long-range support: LiDAR ahead in every eligible scene

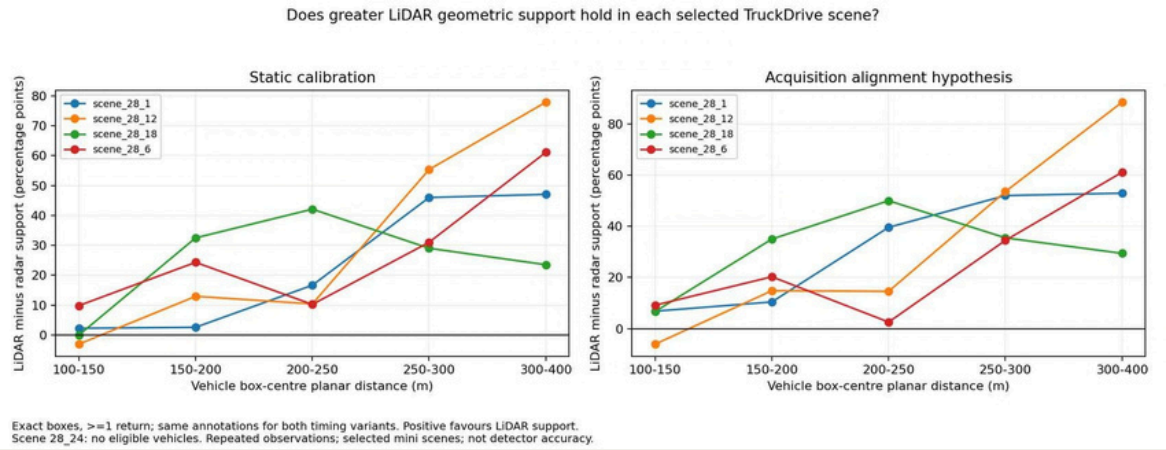


Figure 11. LiDAR minus radar support on labelled vehicles, per scene and distance band

Dataset: TruckDrive mini
Sample: 5 selected scenes (scene_28_1, _6, _12, _18, _24), 190 matched frames; scene_28_24 has no eligible vehicles
Sensor: Paired 4D radar and long-range LiDAR; static calibration (left) and an acquisition alignment hypothesis (right)
Ground truth / prediction: Difference in percentage points between the share of vehicle boxes with ⩾1 LiDAR return and ⩾1 radar return. No model
Limitation: Observations repeat across frames and tracks; five selected scenes are not a population. The density control in Section 4.2 has not been run on TruckDrive

This is the principal long-range outcome of the project. At 200-400m, in 504 vehicle observations LiDAR achieved at least one return within 80.36% of labelled vehicle boxes and radar within 37.50% (static calibration). LiDAR dominates at 200m+ in all four eligible scenes under both timing assumptions; below 150m the difference becomes small and occasionally reversed. Equal weighting of scenes gives LiDAR a 38.17% point advantage (static) and 44.09% (aligned), and removing any single scene leaves LiDAR with at least 31.85 and 37.75%.

Hold-out check. Two clips (28_3 and 28_15) not previously used were sampled at timestamps chosen before any sensor scoring. At 200-400m there are 36 vehicle observations on 18 tracks. LiDAR achieved 86.11% support and radar 33.33% under alignment hypothesis (static: 80.56% and 36.11%), LiDAR ahead in both clips. A single radar-only observation raised support from either sensor to 88.89%, showing that radar sometimes provides evidence not available to LiDAR.

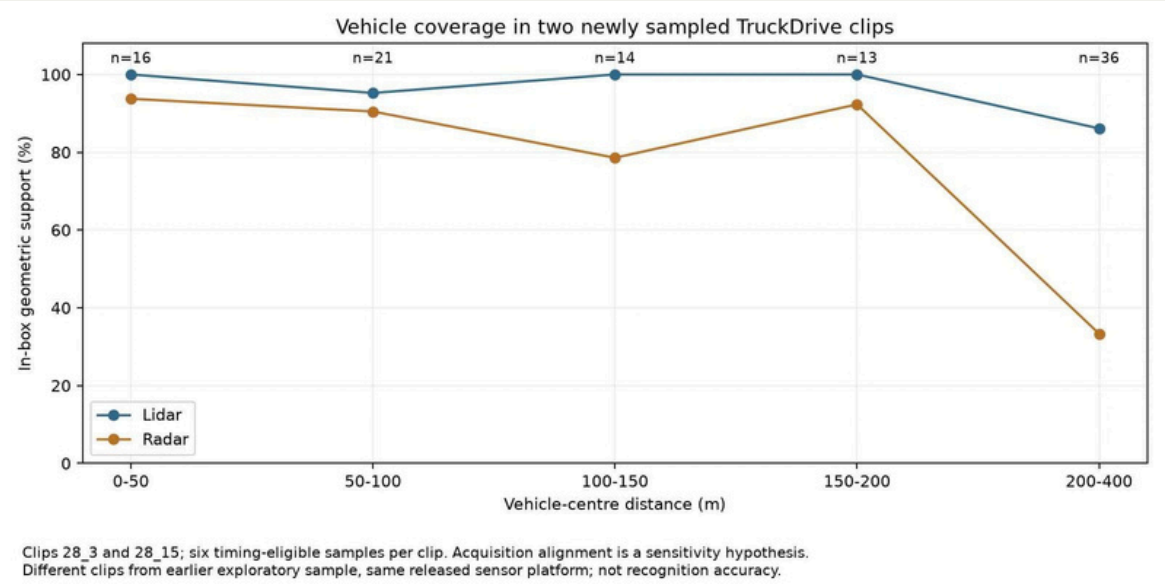


Figure 12.Vehicle coverage in two newly sampled TruckDrive clips

Dataset: TruckDrive mini
Sample: Clips 28_3 and 28_15; six timing-eligible samples per clip, 115 valid object observations
Sensor: Paired LiDAR and radar; acquisition-alignment variant shown
Ground truth / prediction: Share of vehicle observations with ⩾1 in-box return, by vehicle-centre distance
Limitation: Small: n=36 at 200–400 m from 18 tracks, and fewer in nearer bands. Same released sensor platform, so not a new sensor population. Not recognition accuracy

5.3 Does the result survive timing and box size?

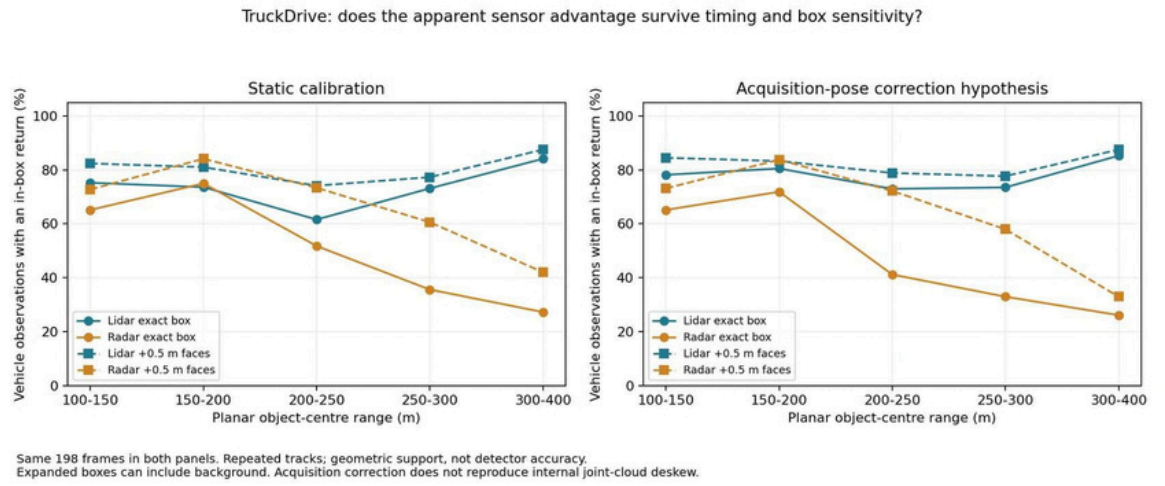


Figure 13.scene_28_1: vehicle support under two timing assumptions and two box sizes

Dataset: TruckDrive mini
Sample: One scene (scene_28_1), the same 198 frames in both panels, vehicle boxes out to 399 m
Sensor: Paired LiDAR and radar; static calibration (left) vs acquisition-pose correction (right); exact boxes vs boxes expanded 0.5 m per face
Ground truth / prediction: Share of vehicle observations with ⩾1 in-box return. No model
Limitation: One scene, few far tracks (five at 250–300 m, three at 300–400 m). Expanded boxes can include background. The correction is a tested hypothesis, not the publisher's own deskew

The far-range pattern holds under every variant: radar's exact-box support falls from about 72–75% at 150–200 m to about a quarter at 300–400 m, while LiDAR stays well above it. The 150–200 m band is where the answer moves: 73.58% LiDAR against 74.96% radar under static calibration, 80.48% against 71.85% after the timing correction, and 83.25% against 83.77% with expanded boxes. No sensor-superiority claim is stable at that range.

5.4 HOW PERSISTENTLY IS A DISTANT VEHICLE SEEN?

In the forward sector (within 15° of the sensor's forward direction), 104 of 107 labelled vehicle tracks had LiDAR support at least once, and 74 had it in three consecutive sampled observations; radar managed 101 and 71. Requiring at least three returns per observation, the persistent counts were 72 for LiDAR and 48 for radar. The first three-sample confirmation happened at 200–400 m for 30 LiDAR tracks and 26 radar tracks with a one-return rule, and for 28 and 4 with a three-return rule. Samples are roughly half a second apart, so a three-sample run takes about one second. The rule is an analysis choice, not a sensor latency or a braking requirement.

5.5 VIEWER REPRODUCIBILITY (GAP G-9)

A second-machine reproduction (EXP-0009) had reported three viewer problems. Kelsey re-tested them on 28 September in the truckdrive_visualizer environment (Python 3.11.11, NumPy 1.26.4, SciPy 1.14.1). This is her verification, not an independent second reproduction.

TABLE 4. TRUCKDRIVE VIEWER ISSUES AFTER RE-TEST

Issue	Status	Effect on analysis
NumPy/SciPy version conflict	Resolved	Both import; pipcheck reports no broken requirements. The earlier conflict was environment-specific; pinning SciPy near 1.14 avoids it
Empty camera channels	Resolved for this local copy	With the corrected folder layout (camera/leopard/‹channel›/images), all 11 Leopard channels hold 259–260 images. The cause was the folder layout, not missing data
2D overlay error (x1 ›= x0)	Not reproduced	The triggering frame was never recorded, so it stays open as unconfirmed rather than fixed
Open3D SetViewPoint() warnings	Still observed	Do not block the 3D or 2D views used in verification

Sources: *evidence/truckdrive/README.md; docs/dataset-surveys/truckdrive.md; docs/truckdrive-multiscene-result.md; docs/long-range-cross-dataset-review.md; experiment-log/0030; docs/final-findings-oct03.md (D01, D03, temporal evidence); result records 0015, 0018, 0020, 0025, 0036, 0038.*

6. STONE AND RADIATE: OFF-ROAD RADAR AND FOG (RICKY YUEN)

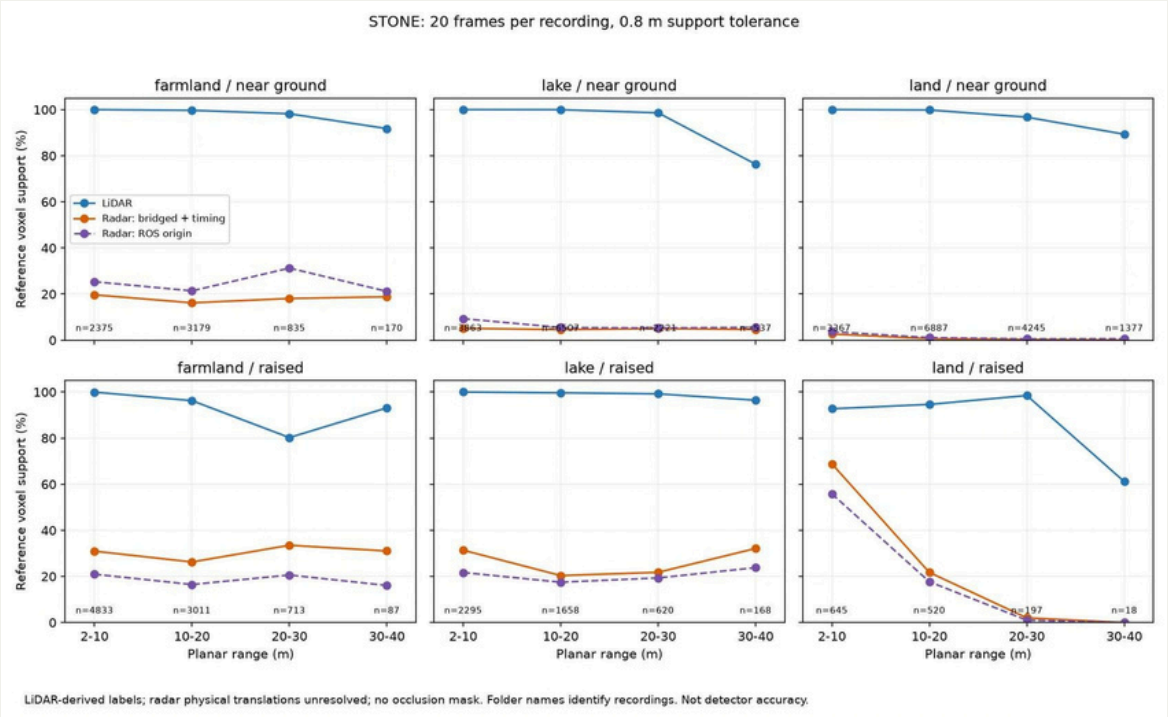


Figure 14. STONE: radar and LiDAR support on off-road terrain

Dataset: STONE (off-road, labelled 4D radar)
Sample: 60 frames: 20 from each of three recordings (farmland, lake, land), 2–40 m
Sensor: 3 Continental ARS 548 4D radars and LiDAR
Ground truth / prediction: Share of LiDAR-derived terrain voxels (near-ground and raised) with a nearby return. No model
Limitation: Released radar mounting translations are zero, so two calibration conventions are shown, and the farmland ordering flips between them. Labels are LiDAR-derived. No occlusion mask

STONE is the only dataset we found that is both off-road and has labelled 4D radar. LiDAR supports nearly all terrain (about 94–100% depending on recording and terrain type). Radar supports little near-ground terrain: 18–24% on farmland, 5–7% at the lake and 1–1.5% on land, across the two calibration conventions. It does better on raised geometry (19–41%).

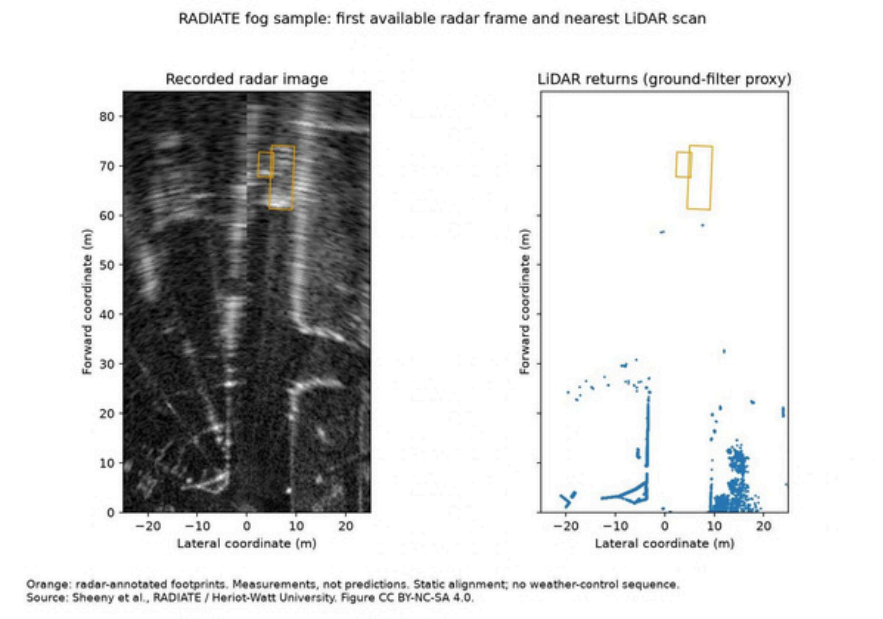


Figure 15. RADIATE fog: radar image and LiDAR returns around labelled vehicles

Dataset: RADIATE public fog sample (Heriot-Watt University)
Sample: fog_6_0: 17 radar frames, 39 vehicle observations from 4 tracks
Sensor: Scanning radar (image) and LiDAR
Ground truth / prediction: Radar-derived vehicle boxes; radar local contrast vs LiDAR returns above a ground proxy. No model
Limitation: One short sequence; labels come from radar, which favours radar-visible targets; no clear-weather control. Figure licensed CC BY-NC-SA 4.0

In fog at 50–75 m, LiDAR had returns above the ground on 0 of 19 labelled vehicle observations, while radar showed local contrast on 16 of 19 under a loose rule and 9 of 19 under the strict rule. Radar contrast is not specific to vehicles, though. Across all ranges, the loose rule also passed 32 of 117 rotated comparison regions that hold no labelled vehicle (against 31 of 39 vehicle footprints); the strict rule passed 8 of 117 (against 17 of 39). Two vehicle footprints held a strict three-sample run and no comparison region did. This is the clearest case in our evidence of radar responding where LiDAR has nothing, but a radar response should not be equated with a recognised target, and these counts are not false-positive rates.

Sources: *docs/stone-environments-followup.md; docs/radiate-fog-pilot.md; docs/final-findings-oct03.md (D04, D23); result records 0028, 0030, 0031.*

7. ADDITIONAL DATASETS REVIEWED

Six further datasets were checkedtosee whether they could answer a question the core datasets cannot.None was run as a detectorbenchmark. Raw files stay outside the repository; pinned sources,sizes and hashes are in therepository manifests.

TABLE 5. ADDITIONAL DATASETS: WHATWAS CHECKED AND WHAT EACH COULD ANSWER

Dataset	What we checked	Question it could answer	Boundary
Boreas	Pinned official documentation; three radar scans and their nearest LiDAR scans decoded	Repeat-route robustness and rolling-scan alignment; independent 2D scanning radar	Decoding, timing and range representation only. N

Dataset	What we checked	Question it could answer	Boundary
WildScenes	Official split files audited; five preselected terrain frames inspected	Independent natural-terrain generalisation from GOOSE	No inference. XYZ only (no intensity); water mapped to an ignored label
RELLIS-3D	Pinned author release metadata	Off-road camera/LiDAR with water, puddle, mud and rubble classes	No raw frames. Velodyne labels transferred from Ouster. CC BY-NC- SA 3.0
Dual-Radar	Official repository and sensor table	Whether two radars (Arbe Phoenix, ARS 548) behave differently on one platform	Source review only
RADIal	Official data and label documentation, processing code	How raw radar processing vs sparse detections changes the evidence	Source review only; different label tasks
View of Delft	Official release and access conditions	Urban radar/LiDAR detection, especially pedestrians and cyclists	Access is request-based and listed for postgraduate and staff researchers; not established for this team

Sources: *docs/additional-dataset-findings-oct01.md; docs/final-findings-oct03.md (D14, D15, additional off-road data); result records 0035, 0039.*

8. THE DETECTOR BENCHMARK: WHAT RAN AND WHAT DID NOT

The comparison that would settle the long-range question needs a radar detector and a LiDAR detector, scored on the same samples and ground truth at 200–400 m. No such pair exists in the releases weinspected. That is a negative finding with a known cause, and the conditional benchmarkplanned for Sprint 3 did not start.

TABLE 6. DETECTOR ATTEMPTS

Attempt	Outcome	Why
L-RadSet PointPillars, radar and LiDAR, on TruckDrive (EXP0018)	Not runnable at range	Both checkpoints crop input at about 70 m; 0 of 504 distant vehicle centres fall inside
Radar detectors on TruckScenes (EXP-0020)	Not runnable	No candidate was both accessible and compatible with the radar point format
CenterPoint LiDAR on TruckScenes (EXP-0019)	Not runnable	Needs spconv, which has no Windows build
PointPillars LiDAR on TruckScenes (EXP-0024)	Ran; low score	mAP 0.1005 with all six LiDARs merged and input rectified upright (car AP 0.41). Trucks score 0.009 despite full coverage

Attempt	Outcome	Why
FCOS3Dcamera on TruckScenes (EXP-0006, EXP- 0010)	Near zero	mAP0.0046 with four cameras, 0.0000 with one,zero-shot

Two routes would unblock a scored comparison: a published radar detector that accepts 200–400 m input, or fine-tuning detectors on truck data (for example, on TruckScenes training data to fix truck and trailer recognition). Training was outside the Sprint 3 plan and would need agreement and a compute plan.

Sources: *docs/detector-compatibility-decision.md; docs/ad-s3-1-detector-feasibility-summary.md; docs/truckscenes-lidar-detector-decision.md; experiment-log/0024-truckscenes-lidar-full-split.md; result records 0021–0023, 0029.*

9. WHAT THE EVIDENCE ADDS UP TO

Each dataset answers a different question. TruckDrive is the only one that can test long range, and there LiDAR keeps more geometric evidence on distant vehicles than radar. TruckScenes has the best matched sensor suite, but its recorded radar stops near 190 m, and its density control shows how much of a support gap can come from point count alone. GOOSE and STONE cover off-road terrain, where radar sees little ground and a terrain model makes errors in both directions. RADIATE shows radar responding in fog where LiDAR has nothing, though not specifically to vehicles.

TABLE 7. DATASET SUITABILITY

Question	GOOSE	TruckScenes	TruckDrive	STONE/RADIATE
Long range (D- 04)?	No: 1.10% of points beyond 150 m	No: recorded radar stops at ~189.5 m	Yes, to ~400 m: LiDAR 80.36% vs radar 37.50% in-box support	No: short-range labels; one fog sequence
Radar vs LiDAR?	No labelled radar	Yes, to ~190 m	Yes, to ~400m	Yes, short range
Off-road terrain?	Yes	No	No	STONE: yes
Weather?	Seasons, rain	One rain and one snow recording	Not analysed	RADIATE: fog
Model result	PTv3, reduced configuration, partial	Camera mAP 0.0046; LiDAR mAP 0.1005	None runnable at range	None

What we can say:

Under the tested protocols, LiDAR put returns on more distant labelled vehicles than radar; on TruckScenes most of that gap is point count; radar coverage varies strongly by object type; terrain models fail in both directions, and map conversion can lose hazards even from correct labels.

What we cannot say:
Which sensor detects better at any range, whether radar is more robust in bad weather, or whether any area is safe to drive.

10. HOW OUR FINDINGS COMPARE WITH INDUSTRY (AIDEN BLAMPAIN)

We checked the team's main findings against what leading autonomous-trucking companies publish: Aurora, Kodiak, Torc, Plus, Waabi and Gatik. Companies publish claims rather than scores on our datasets, so no number on one side is compared with a number on the other. The one measured source is the TruckDrive dataset paper (Ghilotti et al., arXiv:2603.02413). Sources were read on 3 October 2026, and we re-checked the paper's abstract on 6 October.

TABLE 8. OURFINDINGS AGAINST PUBLISHED INDUSTRY POSITIONS

Our finding	What industry says	Verdict
At 200–400m, LiDAR covered distant vehicles far more completelythan radar	Long-range perception is anchored on long- range FMCW LiDAR at 450–500 m (Aurora, Torc with Aeva); radar is the complement	Agrees
No available model can test recognition at 200–400 m	The TruckDrive paper: current models "do not generalize to ranges beyond 150 meters". Company horizons of 450–1,000 m come without published benchmarks	Agrees (measured); company claims cannot be checked
No radar-first detector to benchmark	Every trucking company whose sensors were found (5 of 6) fuses LiDAR, radar, and cameras	Agrees
4D radar is on our datasets' trucks but has no public models	4D or imaging radar is on Kodiak (since 2021), Aurora and Plus trucks	Agrees
Radar covers people, cones and signs less consistently than LiDAR, but adds some evidence	Radar is used for velocity, weather and long range; cameras and LiDAR handle small objects and semantics	Agrees
Fog hints radar helps where LiDAR is sparse; LiDAR kept strong car coverage in our rain and snow recordings	Radar is presented as the adverse-weather sensor; weather limited Aurora's driverless operations about 40% of the time in 2025	Tension
Public models transfer poorly to a new truck unless the sensor frame is corrected; heavy trucks stay unrecognised	Waabi reports moving its stack to a differently sensored truck with no retraining, without published metrics	Tension (unverified claim)
Off-road terrain models confuse rocks, tall grass and walls with ground	Kodiak runs driverless trucks on oilfield lease roads but publishes nothing on terrain perception	Cannot tell

Our long-range result agrees with the industry's sensor choices and with the TruckDrive authors' own research, while the company's range claims of 450–1,000 m are not backed by published benchmarks. The two tensions are the most useful open questions for the client: whether radar really covers bad weather, which should be tested per condition rather than assumed, and whether models can move between trucks.

Sources: *docs/ad-3-industry-check.md (sources A1–A3, K1–K3, T1–T3, P1, W1–W2, G1, M1); docs/domain-study/README.md.*

11. LIMITATIONS, OPEN ITEMS AND DECISIONS REQUESTED

TABLE 9. OPEN ITEMS

Item	Status	What would close it
Independent review of recent analyses	The range-band and density-matched analyses (EXP-0033, EXP-0034) are logged as agent-assisted and not yet human-accepted. The TruckDrive survey is marked not yet reviewed. The AD-3 industry check has no assigned reviewer	Named reviewers signoff in the repository
Outside-reader cold reads (P-6, DS-4)	Not yet performed by anyone outside the author pairs	Two readers follow docs/cold-readprotocol.md; findings recorded verbatim
Regeneration from raw data by an outside member (A-6)	Not performed for the range-band results as of 29 September	A teammate with the TruckScenes mini release reruns the acceptance check
Density control on TruckDrive	Not run; the TruckDrive long-range gap is not yet separated from point count	Rerun record 0041's method on the TruckDrive paired evidence
GOOSE frame count (G-2)	The archive holds 961 frames, one more than the published split of 960	Identify the extra frame against the publisher's list
TruckScenes radar boundary (G-10)	Cause of the ~189.5 m limit unknown	Vendor or publisher documentation
TruckDrive overlay error (G-9)	Not reproduced; triggering frame unknown	Record the frame if it recurs
TruckDrive licence (G6)	Non-commercial terms and an autonomous-vehicle business restriction sit against the project's Creative Commons / open-source deliverable	Client decision
Scope and headline question	Client confirmation of the revised scope and of the D-04 question is not recorded in the repository	Client and mentor confirmation

DECISIONS WE WOULD VALUE THE CLIENT'S VIEW ON

1. Confirm the D-04 headline question and the reporting range bands with Fabian.
2. Decide how TruckDrive-derived material may be shared, given its licence terms (G-6).
3. Decide whether a future team should fine-tune a LiDAR detector on truck data and run the density control on TruckDrive. Both need compute and agreement. 4. Set the priority between weather (matched clear and adverse recordings from the intended sensor platform) and off-road perception (camera texture for surface categories, and water, mud and rock examples).

12. REPRODUCIBILITY AND HANDOVER

Everything in this report can be traced and, where the raw data is available, rerun.

- Collated evidence: evidence/GOOSE/ (README, manifest, and SHA-256 verification of every copied file) and evidence/truckdrive/ (README and dataset statistics). TruckScenes evidence is under docs/evidence/truckscenes/ and TruckScenes - Fatima/.
- Results store: 41 schema-valid result records in results/records/, each stating dataset, sensors, annotation schema, task type, environment, commands, and evidence. Failed and partial runs are recorded too.
- Experiment logs: 34 entries in experiment-log/ with goals, environments, commands, outcomes, and attempted fixes.
- Checks: python scripts/validate_result.py and python scripts/validate_experiment_logs.py, plus the test suite, run in CI on Linux and Windows.
- Raw data stays outside the repository; manifests record sources, revisions, and hashes.

SECOND-MACHINE REPRODUCTION CHECK

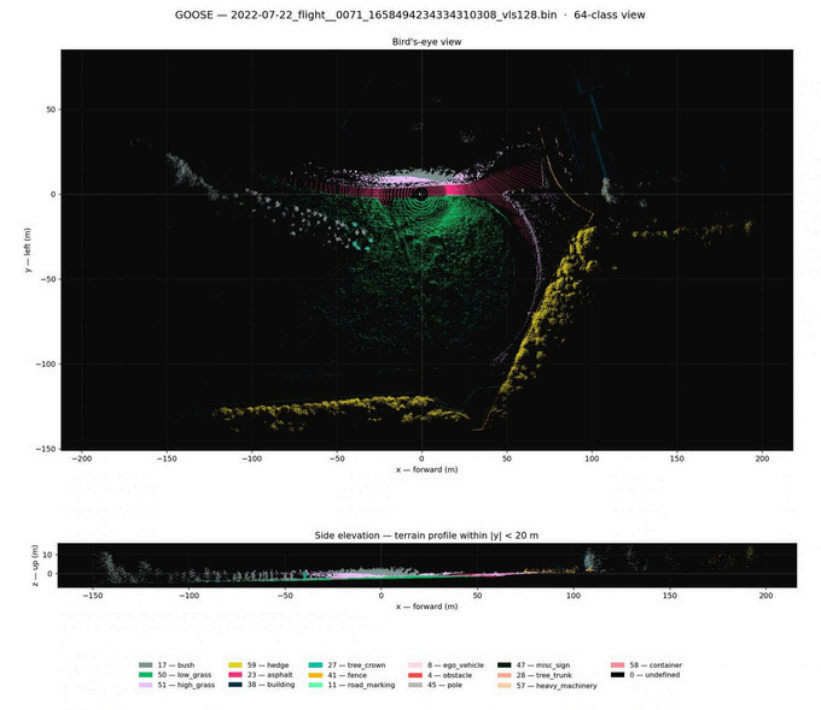


Figure 16.GOOSE smoketestreproduced on asecond machine

Dataset: GOOSE 3D, base validation split
Sample: Frame index 0 (2022-07-22_flight, 169,883 points)
Sensor: LiDAR only
Ground truth / prediction: Human labels, rendered by scripts/goose_render_frame.py
Limitation: Shows the setup and renderer are repeatable; no model was rerun

On 29 September 2026 Fariya followed Ricky's GOOSE Windows setup (EXP-0001) on a second machine, as a member outside the GOOSE pair. The fresh download held 961 LiDAR scans and 961 label files, and frame 0 rendered with the same 169,883 points and class breakdown as the original run (bush 35.87%, low grass 20.61%). The pinned packages worked on Python 3.12 as well as the original 3.11.
Outcome: pass.

APPENDIX A: FINDINGS REGISTER

The project's twenty findings (docs/final-findings-oct03.md). Identifiers are stable; D11, D16 and D17 were merged into other findings. Support means at least one return inside a labelled box, not recognition.

ID	Finding	Key evidence
D01	LiDARsupplied data from distant TruckDrive vehicles moreoften than radar	200–400 m, hold-out clips: 31/36 vs 12/36
D02	Strongradar coverage of nearby trucks did not carry overtopeople and roadwork objects	0–25 m: trucks 99.12%, adults 67.66%, cones 39.00%
D03	Radarsometimes supplied object evidence where LiDARhad none	Union rises 86.11% to 88.89%
D04	Weathersamples do not establish a sensor that wins in every condition	Fog 9/19 strict radar vs 0/19 LiDAR; rain and snow LiDAR ⩾95%
D05	LiDAR returned terrain points the model sometimes labelled incorrectly	Tall grass 48.94% called ground
D06	Calling a surface ground did not establish its surface category	Gravel 99.82% ground, 80.07% correct category
D07	More context improved most terrain examples but cost more and worsened one	6 of 7 improved; ~28% more time
D08	A strong overall terrain score hid a weak result in another recording	Soil 99.53% pooled vs 7.92% rainy
D09	Correcting sensor timing changed how much evidence appeared missing	Radar-only 185 to 39
D10	The inspected software could not compare recognition of distant vehicles	Input crop ~70 m
D12	The terrain model also rejects some candidate ground obstacles	672 of 4,232 cells (15.88%)
D13	A majority-ground map can hide a hazard even with correct labels	as 2,853 of 1,312,878 points
D14	Important off-road conditions remain unchecked, including water	No water points in selected frames
D15	Independent terrain data needs an input compatibility check first	WildScenes XYZ vs GOOSE XYZ + intensity
D18	Nearby maps contained substantial gaps with no recorded points	43.46% of half-metre cells
D19	Ground-labelled patches had different local shapes	64,920 points ⩾2 m above ground reference
D20	Obstacle returns occurred at different vertical levels	28,958 of 70,807 cells met plane-fit criteria

ID	Finding	Key evidence
D21	A widerfootprint check removed many positions that passedasingle-cell check	2.5 m squarekept 23.48%
D22	Poor camera visibility did not automatically mean missing LiDAR evidence	95.14% LiDAR, 27.03% radar
D23	Radar contrast appeared in labelled vehicles and in unlabelled comparison regions	Loose rule: 31/39 targets, 32/117 comparisons

APPENDIX B: SOURCES

Project repository (all paths relative to the repository root, branch main, commit 1704678):
•Synthesis:docs/final-findings-oct03.md,docs/dataset-suitability.md,docs/client-comparisonbrief.md
• GOOSE: evidence/GOOSE/, docs/goose-multiscenario-terrain.md, docs/goose-error-cases.md, docs/goose-failure-causes.md
• TruckScenes: TruckScenes - Fatima/RANGE-BANDS.md, docs/truckscenes-rawcomparison.md, docs/long-range-evidence-audit.md, docs/evidence/truckscenes/densitymatched/, experiment-log/0024-truckscenes-lidar-full-split.md
• TruckDrive: evidence/truckdrive/, docs/dataset-surveys/truckdrive.md, docs/truckdrivemultiscene-result.md, docs/long-range-cross-dataset-review.md
• STONE and RADIATE: docs/stone-environments-followup.md, docs/radiate-fog-pilot.md
• Additional datasets: docs/additional-dataset-findings-oct01.md
• Detectors: docs/detector-compatibility-decision.md,docs/ad-s3-1-detector-feasibilitysummary.md
• Industry: docs/ad-3-industry-check.md

EXTERNAL

• Ghilotti et al., TruckDrive: Long-Range Autonomous Highway Driving Dataset, arXiv:2603.02413 (March 2026).
• Dataset authors and publishers: GOOSE (Fraunhofer IOSB), MAN TruckScenes, TruckDrive (Torc Robotics), STONE, RADIATE (Heriot-Watt University), Boreas, CSIRO WildScenes, RELLIS-3D, Dual-Radar, RADIal, View of Delft.
• Company sources for Section 10 are listed with dates and links in docs/ad-3-industry-check.md.